



Original Article

Carrageenan-grafted-poly(acrylamide) magnetic nanocomposite modified with graphene oxide for ciprofloxacin removal from polluted water

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ABSTRACT

The κ -carrageenan-grafted-poly(acrylamide)/Fe₃O₄-GO (κ -car-g-PAAm@Fe₃O₄-GO) adsorbent was attained in three stages, utilizing grafted poly(acrylamide) (PAAm) onto κ -carrageenan (κ -car) in Fe₃O₄ and graphene oxide (GO) presence. Various characteristic analyses were carried out to peruse the chemical properties, morphological features, thermal stability, magnetic behavior, and porosity of the prepared nanoabsorbent. According to the results obtained from the analyses, the prepared nanoabsorbent showed superparamagnetic behavior. It rendered a semicrystalline structure by Fe₃O₄ presence in the amorphous matrix of the κ -car-g-PAAm. The SEM images exhibit well dispersion of the Fe₃O₄ magnetic nanoparticles (MNPs) and lamellar GO sheets throughout the hydrogel context. Besides, along with the GO incorporation into the structure, the BET surface area (2.361 m²/g), pore volume (0.008 cm³/g), and pore size (137.49 nm) of the κ -car-g-PAAm@Fe₃O₄-GO nanoabsorbent were higher than κ -car-g-PAAm@Fe₃O₄. The maximum adsorption capacity (Q_{max}) at an ambient temperature for CPFX adsorption was 1146.467 mg/g. These adsorption experiments' data were described by the Freundlich isotherm, and the pseudo-second-order explains the adsorption kinetics well. The retrievability of the prepared nanoabsorbent was studied and indicated that it could be isolated effectively and reused in three consecutive cycles without remarkable adsorption efficiency loss.

1. Introduction

The water resource shortage is a significant problem that threatens societies these days. The accessible water resources are deteriorated due to the activity and progress of the industries, mismanagement of the urban waste, and taking no account of managing the water resources [1–5]. The extensive development and application of antibiotics in treating bacterial infections of animals and humans. Despite categorizing antibiotics as crucial medical substances, the abandonment of these materials into the environment and aqua resources has caused severe consequences even at lower dosages [6–9]. In this regard, quinolone antibiotics, i.e., ciprofloxacin (CPF), levofloxacin, and pefloxacin, are not metabolically stable and discharge several fundamental chemical substances into the environment [10,11]. Releasing antibiotics in water sources has harmful repercussions, such as vomiting, diarrhea, allergies, etc. In addition, antibiotic deposits have noticeable disadvantages in algae growth rate and water species photosynthesis [12–15]. Releasing

antibiotics in the ecosystem also elevates bacterial infection resistance to several employed medicines [16,17]. CPF, a second-generation fluoroquinolone, has been prescribed for treating diverse infections, namely, urinary infections, bone infections, prostatitis, etc. CPF is an affordable drug in various consumable forms, including oral pills, eye drops, hydrogels, etc. [18,19]. This antibiotic is a dangerous pollutant causing irrecoverable impacts in aqueous systems; hence, it seems crucial to design and employ an effective technique for CPF removal from the water resources [20,21].

Eliminating the released quinolone antibiotics in the aqueous media has been executed through different routes, like biological procedure, ion exchange, osmosis, and adsorption. The adsorption approach is the most conventionally employed approach, according to the beneficial features, such as using fewer chemicals and organic solvents and providing a wide spectrum of concentrations, turning it into an effective and eco-friendly solution [22]. Therefore, several micro and nano-scale structures were suggested as appropriate adsorbents for an effective

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CPFX elimination from water [23,24]. Natural polymer-based composites have been used as a biodegradable, available, cost-effective, and less toxic adsorbent in adsorption systems [25–27]. Many of the natural polymeric hydrogels comprised of xanthan gum, cellulose, ghatti gum, karaya gum, etc., can efficiently capture the pollutants in their three-dimensional (3D) structure, leading to adsorption efficacy enhancement by upgrading the interactions between adsorbent's superficial functional groups and contaminant molecules [28]. Composite hydrogel is defined as a category of hydrogels that are prepared through grafting the hydrophilic functional groups, including carboxylic acid, amine, sulfuric acid, amide, imide, and hydroxyl to the backbone of polymeric string, and incorporating additives like nanoparticles (NPs), graphene-based substances, MOFs, etc. [29–32]. Carrageenans is a natural polysaccharide with a linear polymeric chain employed as the gelling agent in various pharmaceutical and food industries.

Nonetheless, remarkable properties, including cost-effectiveness and sustainability, have developed κ -car utilization in hydrogel-based adsorbent systems [33,34]. On the other hand, the κ -car polymeric chain's functional groups, including sulfate and hydroxyl, render a substrate to modify conveniently by graft copolymerization and applying a cross-linker or interacting with various nanomaterial's type through hydrogen bond networks and electrostatic interactions [35–37]. Thus, the antibiotic adsorption capability of hydrogels containing natural polymers from water arose from polymeric string's reactive hydrophilic functional groups like amide, hydroxyl, etc. In contrast, their persistence toward dissolution is attributed to the cross-linking between the polymeric chain's network. Besides, nanomaterials as additives provide an improved surface area and retrievability, leading to an enhanced adsorption capacity.

Due to the increasing demand for the reduction of NPs loss in their recovery from the reaction mixture, it is preferred to apply magnetic composites or inherently magnetic substances as adsorbents [38,39]. With this concern, the magnetic nanocomposite renders suitable and convenient systems for water decontamination and contaminant adsorption facets [40]. Graphene oxide (GO) nanosheets are great additives for enhancing hydrogel-based adsorbents due to their abundant hydrophilic functional groups and surface epoxide, carboxyl, and hydroxyl functional groups [41,42]. The GO sheets' colloidal stability and hydrophilic nature are higher than graphene layers, which correspond to the COO^- anions on the GO's surface. Additionally, their physicochemical interactions via uncharged epoxide, OH^- , and π - π interactions confirmed the GO sheets' amphiphilic nature. Since the specific surface area of the GO sheets is high, incorporating these substances in the adsorbent's structure results in a massive increment in the composite's surface area, porosity, and adsorption efficiency [43,44]. GO can provide a fast interfacial connection with the hydrophilic substrate by chemical or hydrogen interactions. According to the remarkable GO sheets' thermal and mechanical persistence, the GO-containing hydrogel nanocomposites' thermal behavior resistance and mechanical strength might be developed [42]. In the current study, To effectively separate CPFX from an aqueous medium, a heterogeneous adsorbent was proposed based on an *in situ* copolymerization. Regarding this, the graft copolymerization of the AAM onto κ -car was exerted in the GO and Fe_3O_4 MNP additives presence. The ammonium persulfate (APS) and N, N-methylene bisacrylamide (MBA) were applied as a redox initiator and covalent cross-linker. The graft copolymerization then separated the GO layers from the inner gallery and led to form GO/polymer nanocomposites. Diverse functional groups in the κ -car-g-PAAm@ Fe_3O_4 -GO nano-adsorbent structure and enhanced porosity could effectively interact and adsorb CPFX antibiotic contaminant from an aqueous medium with an increased adsorption capacity.

2. Experimental

2.1. Materials and instruments

All of the materials and apparatus used in this work have been summarized in Table 1 and Table 2.

2.2. Fabrication of nanosorbent

2.2.1. Preparation of the graphene oxide (GO)

GO was prepared by Hummer's process from graphite [45]. In summary, 1.0 g of layered graphite, 0.5 g of sodium nitrate, and 23.0 mL of concentrated sulfuric acid were poured into a 1.0 L beaker. Then, the reaction mixture was sonicated in an ultrasonic bath at 0 °C. Next, 3.5 g of potassium permanganate was gradually added to the mixture. The reaction temperature should not exceed 20 °C during the gradual increase of potassium permanganate. After that, H_2O was added to the reaction, and the reaction temperature was increased to 98 °C. This temperature was maintained for 30 min to complete the reaction process. After that, the beaker was placed in an ice bath for 20 min to reduce its temperature. Next, 500.0 mL of H_2O and 12.0 mL of H_2O_2 (30%) were added to increase the reaction temperature. After adding hydrogen peroxide, the mixture quickly altered from dark brown to light yellow, indicating GO production. After cooling at 25 °C, the prepared product was isolated through decantation by washing several times with distilled water until reaching 5 to 6 pH. Ultimately, the sample was dried in an oven, and the GO was prepared (Fig. 1) [46].

2.2.2. Preparation of the Fe_3O_4

The iron oxide was synthesized as follows [47]. In this line, $\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$, $\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$, and H_2O were stirred under N_2 gas at 80 °C. Then, NH_4OH was added to the above solution to form Fe_3O_4 MNPs. Afterward, the black sediment was collected with a magnet, cleaned numerous times with H_2O , and dried at 70–80 °C (Fig. 2).

2.2.3. Preparation of the κ -car-g-PAAm@ Fe_3O_4 -GO nano-adsorbent

The GO decorated κ -car-g-PAAm@ Fe_3O_4 as a GO-based polymer composite was fabricated by an *in situ* copolymerization approach (Fig. 3). In the *in situ* copolymerization, the GO is mixed with a liquid polysaccharide, initiator, and monomer. In this regard, APS as a redox initiator and MBA as a cross-linker were applied. The nanocomposite was fabricated at optimum conditions as follows: 400.0 mg of obtained GO was dispersed entirely in 20.0 mL of H_2O for 30 min. Then, dissolved κ -car (1.0 g of 50.0 mL of H_2O) was placed under a nitrogen atmosphere for 5 min. Afterward, dispersed GO was added. Next, APS (0.228 g) was added to the mixture. After 45 min, 3.0 g of AAM monomer was dissolved in 25.0 mL of H_2O and added to the reaction mixture slowly. After 20 min, 1.0 g of the as-prepared Fe_3O_4 MNPs was added and stirred for

Table 1
Substances, brands, and purity used in this study.

Materials	Brand and purity
$\text{FeCl}_3 \cdot 6\text{H}_2\text{O}$	Merck (Code No. 10025–77-1)
$\text{FeCl}_2 \cdot 4\text{H}_2\text{O}$	Merck (Code No. 13478–10-9)
κ -car	Sigma Aldrich (Code No. 11114–20-8)
Layered graphite	Sigma Aldrich (Code No. 902926)
Sodium nitrate	Sigma Aldrich ($\geq 99.0\%$)
Potassium permanganate	Sigma Aldrich ($\geq 99.0\%$)
Ammonium persulfate (APS)	Sigma Aldrich ($\geq 98.0\%$)
Acrylamide (AAM)	Sigma Aldrich ($\geq 99.0\%$)
N-N-Methylenebisacrylamide (MBA)	Sigma Aldrich ($\geq 99.5\%$)
Ciprofloxacin (CPFX)	Sigma Aldrich ($\geq 98.0\%$)
Hydrogen peroxide	Merck (30.0 %)
Sulfuric acid	Merck (98.0 %)
Hydrochloric Acid	Merck (37.0 %)
Ethanol	Sigma Aldrich (96.0 %)
Ammonia	Merck (25.0 %)

Table 2

Apparatus names and models used in this study.

Equipment	Brand and model
FTIR	Shimadzu FTIR-8400S
EDX	VEGA-TESCAN-XMU
XRD	DRON-8 X-ray diffractometer
FESEM	Hitachi S-5200 and ZEISS SIGMA
TGA	TGA-Bahr-STA 504, under argon atmosphere
BET	2020 ASAP TM micromeritics
AFM	Ara research multi model
UV-vis (solid phase)	Shimadzu-UV-2550/220v
Ultrasound bath	KQ-250 DE
Filtration paper	Whatman (grade 602 h, Particle retention < 2 μm)
Autoclave	Reyhan Teb, 2KW-220v

15 min. Then, 1.6 g of MBA was added, and the reaction proceeded at 60 °C for 4 h to complete the polymerization. After 4 h, the reaction mixture was added to a flask containing 250.0 mL of ethanol to obtain the κ -car-g-PAAm@Fe₃O₄-GO precipitate. Eventually, the precipitate was collected magnetically, alternatively rinsed with a mixture of water and ethanol (2:1), and the prepared κ -car-g-PAAm@Fe₃O₄-GO nano-adsorbent was dried for 17 h.

2.3. Batch experiments

The batch experiments of the κ -car-g-PAAm@Fe₃O₄-GO nano-adsorbent in 10 mL of aqueous solution were investigated by CPFX removal [48]. Experimental successions have been executed at a 4–9 pH

range, 3–20 mg adsorbent dosages, 50–500 ppm initial concentration, and a reaction time of 5–45 min to peruse the effect of the pH, adsorbent quantity, preliminary concentration of the CPFX, and interaction time. The HCl and NaOH were employed to regulate the pH of the solution in the range of 4 to 9. The concentration of the CPFX in an aqueous solution was monitored via UV-vis spectroscopy. Besides, the removal efficiency percentage of every experiment and the adsorption capacity (Q_e) were calculated via Eq.1 and Eq.2, respectively [49].

$$R(\%) = \frac{(C_0 - C_i)}{C_0} \times 100 \quad (1)$$

$$Q_e = \frac{(C_0 - C_i) \times V}{m} \quad (2)$$

where C_0 (ppm) is the initial concentration of the ciprofloxacin in the solution, C_i stands for the residual CPFX concentration in every experimental sequence, m (g) is the mass of the applied adsorbent, and V (L) demonstrates the volume of the antibiotic solution.

2.4. Regeneration and retrievability

For the evaluation of the applied CPFX-loaded κ -car-g-PAAm@-Fe₃O₄-GO nanoadsorbent's retrievability, it was immersed in ethanol and vigorously stirred at an ambient temperature for 4 h to desorb into the elution solution. After that, the nanoadsorbent was magnetically collected, rinsed with H₂O, and dried for further experiments. The concentration of desorbed antibiotic in the elution solution was

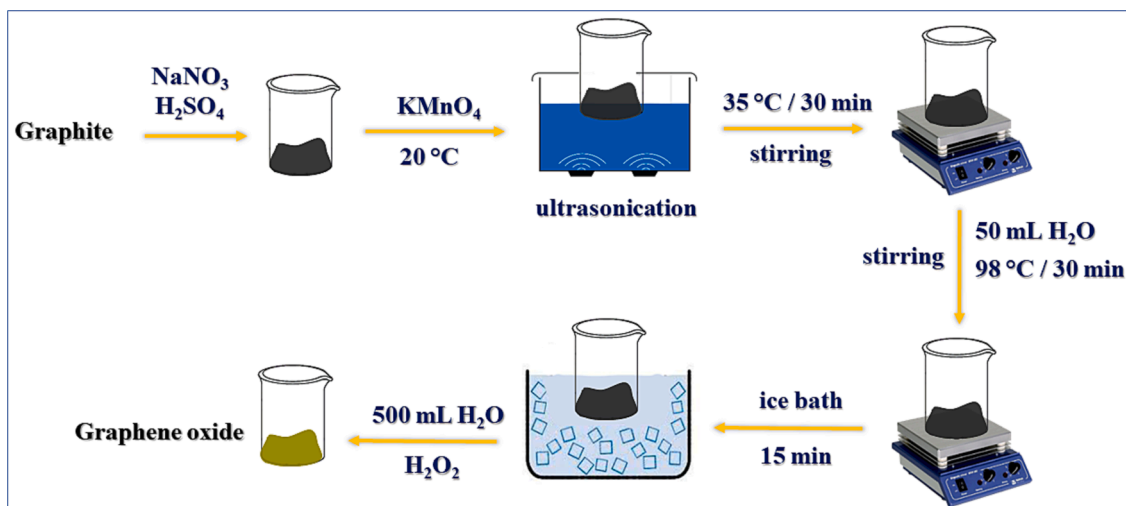


Fig. 1. The schematic representation of the GO preparation.

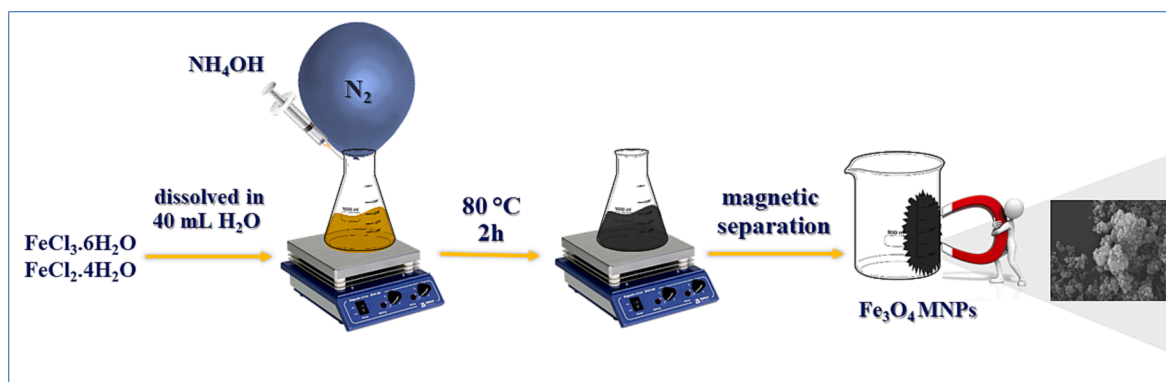


Fig. 2. The schematic of Fe₃O₄ MNPs preparation.

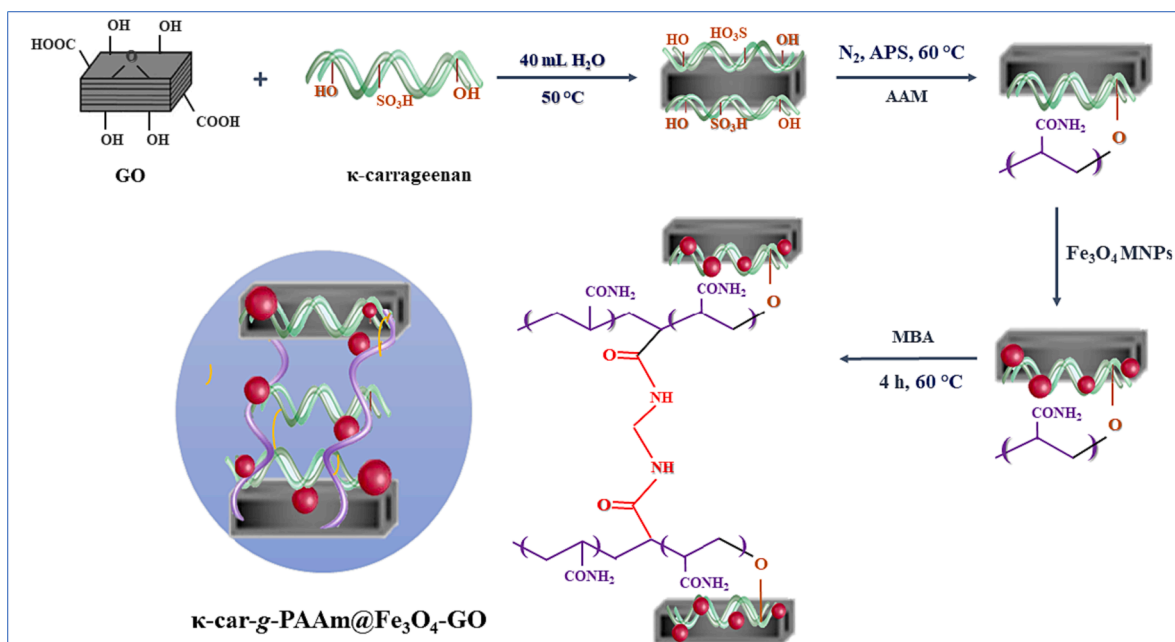


Fig. 3. The schematic illustration of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent.

measured by a UV–visible spectrophotometer. The reusability of the nanoadsorbent was perused via consecutive adsorption–desorption runs. Desorption percentage (D%) was computed by applying Eq.3 as follows:

$$D(\%) = \frac{A}{B} \times 100 \quad (3)$$

Where A (mg) represents the desorbed contaminant's amount in the washing solution, and B (mg) is the adsorbed contaminant's amount on the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent.

3. Results and discussion

Due to the three general preparation procedures for the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent (such as GO, Fe₃O₄ MNPs, and κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent preparation), versatile analytical analyses were provided to characterize various feature of the prepared nanoadsorbent, which will be discussed.

3.1. Preparation of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent

The prepared nanocomposite was prepared and exerted for CPFX removal from an aqueous medium. As depicted in Fig. 3, preparing the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent has different stages. κ -car gum is a natural polysaccharide and has hydroxyl groups in its structure. Nevertheless, utilizing individual κ -car is unfavorable based on its low adsorption capacity and meager thermal resistance. Thus, to overcome this problem, a novel κ -car-based nanocomposite hydrogel was prepared via graft copolymerization with AAM in the presence of GO and using MBA cross-linker to create a three-dimensional (3D) hydrogel network in parallel of addition of Fe₃O₄ MNPs. The GO was incorporated into the nanocomposite hydrogel to give a high specific surface area and more reactive functional groups to the nanocomposite, and Fe₃O₄ MNPs were added to the nanoadsorbent to be recycled conveniently. Hence, the prepared nanoadsorbent with increased stability in an aqueous medium and numerous functional groups possesses a desirable performance in CPFX adsorption at neutral pH.

3.2. Characterization of the prepared κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent

3.2.1. FTIR spectroscopy

FTIR was utilized to investigate the step-by-step preparation of κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent and to approve the presence of new functional groups in each preparation stage. As demonstrated in Fig. 4a, the prominent peak at 580 cm^{−1} is related to the Fe–O bond, confirming the Fe₃O₄ MNPs preparation [39]. Also, the peak in the area of 3400 cm^{−1} is related to the O–H stretching vibration in the structure of Fe₃O₄ MNPs [50]. The infrared spectrum of pure κ -car gum is displayed in Fig. 4b. The broad peak in the 3400 cm^{−1} is assigned to the hydroxyl group. The 980–1084 cm^{−1} peak shows C–O–C assigned to the cyclic ether. The emerged band in 2928–2932 cm^{−1} corresponds to C–H stretching vibration. Furthermore, S = O sulfate stretching vibrations arose at 1225 cm^{−1} [51]. The reaction of κ -car gum with APS as a radical initiator produced hydroxyl radical species, which makes the κ -car gum

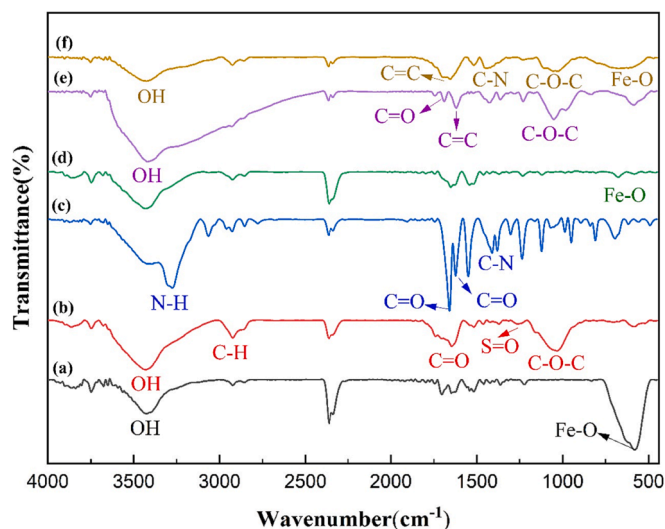


Fig. 4. FTIR analysis of (a) Fe₃O₄ MNPs, (b) κ -car gum, (c) κ -car-g-PAAm, (d) κ -car-g-PAAm@Fe₃O₄, (e) GO, (f) κ -car-g-PAAm@Fe₃O₄-GO.

react with other substances to perform physicochemical reactions. In the next stage, the AAM monomer was used for copolymerization accomplishment. Then, the MBA cross-linker was applied to cross-link the polymeric chains via PAAM double bonds. This copolymerization makes pure κ -car gum more active and reactive regarding functional groups. Fig. 4c, representing the κ -car-g-PAAM FTIR spectrum, shows a peak in the 3000–3500 cm^{-1} region related to N–H vibrations in the structure of AAM and hydroxyl groups in κ -car gum. Also, the peak at 1412 cm^{-1} and 1670 cm^{-1} is related to the stretching vibrations of AAM's C–N and carbonyl groups, respectively. A peak at 1620 cm^{-1} indicates the presence of C = O groups of the amide group (type II) in the cross-linking structure of MBA [52]. In this step, the *ex-situ* prepared Fe_3O_4 MNPs were added to the κ -car-g-PAAM, which was authenticated with a peak at 580 cm^{-1} in Fig. 4d [53,54]. Fig. 4e is attributed to GO. There are four

characteristic absorption bands in this spectrum; first, the peak at 1062 cm^{-1} is assigned to the bending vibration of the C–O–C bond, and at 1565 cm^{-1} , the C = C vibrations of the benzene rings forming the structure of GO can be observed [55]. Besides, the absorption band of the carbonyl group also appears at 1724 cm^{-1} , and the peak with a high intensity related to the stretching vibration of the O–H bond is observed at 3400 cm^{-1} [38,56]. Finally, Fig. 4f represents the κ -car-g-PAAM@- Fe_3O_4 -GO nanoadsorbent's FTIR spectrum, prepared by adding GO via an *in situ* route. This nanocomposite has characteristic peaks, like a medium-intensity peak at 1652 cm^{-1} , confirming the C = C stretching vibration of benzene rings of the GO. The presence of the absorption band at 1430 cm^{-1} demonstrates the stretching vibration of the C–N group in the AAM monomer [57]. Moreover, the peak at 3420 cm^{-1} specifies the O–H groups in water. All the FTIR spectra are brought

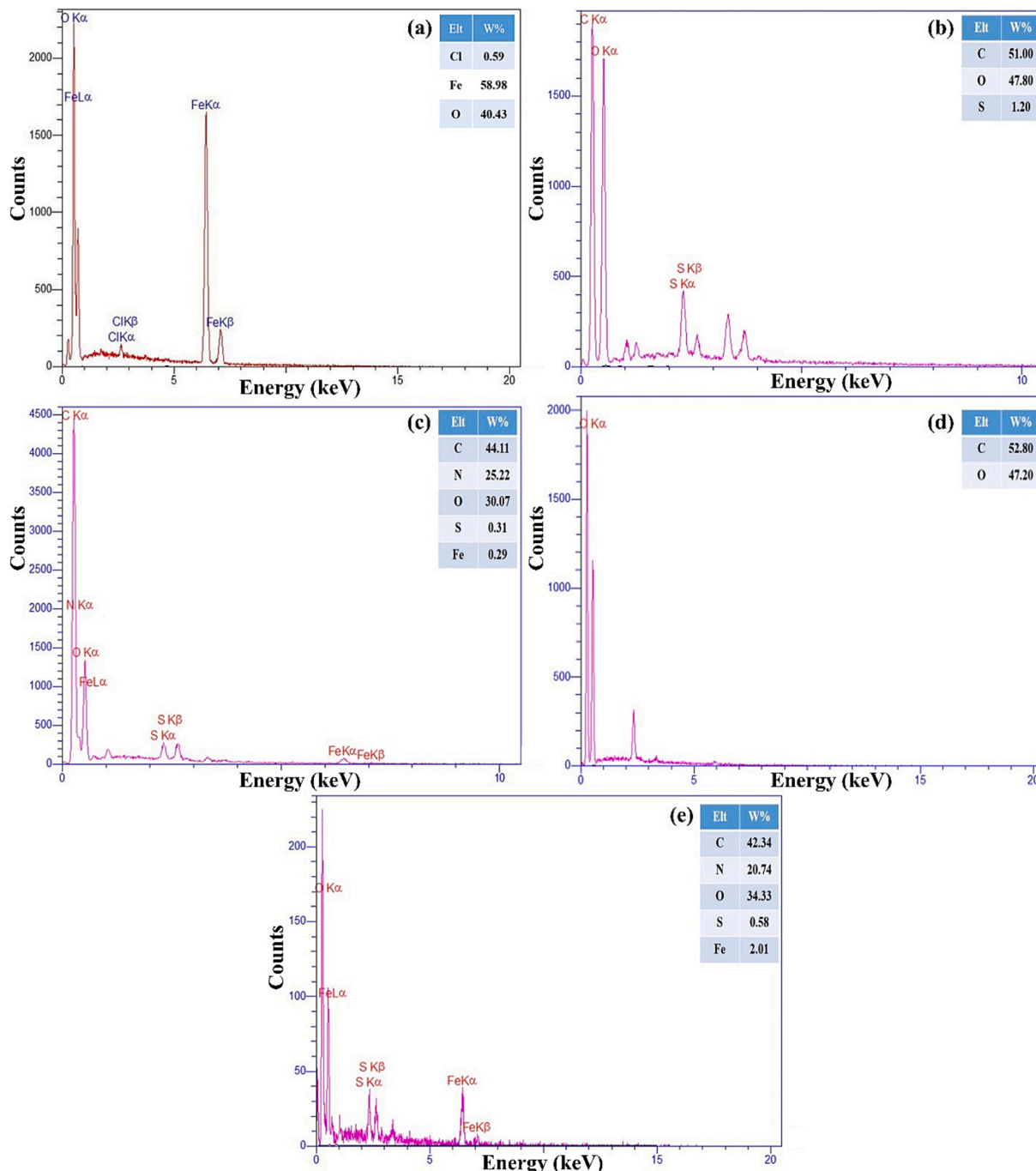


Fig. 5. EDX analysis of (a) Fe_3O_4 MNPs, (b) κ -car gum, (c) κ -car-g-PAAM@ Fe_3O_4 , (d) GO, (e) κ -car-g-PAAM@ Fe_3O_4 -GO.

together so that the changes that occur during the final nanocomposite preparation can be clearly seen and defined.

3.2.2. EDX spectroscopy

Detection and identification of elements in the samples were performed using the qualitative analysis method obtained from EDX. Thus, the constituent elements of the prepared samples can be examined in the EDX spectrum. As shown in Fig. 5a, the EDX spectrum of Fe_3O_4 MNPs has been confirmed with Fe and O elements. Fig. 5b displays the EDX spectrum of pure κ -car gum with high-intensity peaks corresponding to C, O, and S elements. Furthermore, multi-step modifications have been carried out on κ -car gum. In addition to κ -car-related elements, nitrogen, and iron peaks indicate the presence of PAAm, MBA, and Fe_3O_4 MNPs in the structure of κ -car-g-PAAm@ Fe_3O_4 (Fig. 5c). According to Fig. 5d, the elements related to GO, which include C and O with 52.80 W% and 47.20 W%, respectively, are authenticated. In Fig. 5e, the elements associated with the κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite were investigated. This diagram also has high-intensity peaks of C, N, O, S, and Fe elements, confirming the nanocomposite formation.

3.2.3. FESEM microscopy

FESEM identifies the morphology of the prepared nanocomposite. As shown in Fig. 6a, the spherical morphology of Fe_3O_4 MNPs can be seen through FESEM images. In addition, due to the strong interparticle Van der Waals force and magnetic attraction between Fe_3O_4 MNPs, some aggregations are observed in the sample [58]. Fig. 6b depicts the morphology of κ -car gum with a uniform structure that includes a polymer matrix [59,60]. Fig. 6c is related to the κ -car-g-PAAm hydrogel structure, which has a swollen spherical topology throughout the κ -car gum matrix, which can be associated with the uniform grafting of PAAm on the κ -car gum's chain. After adding spherical Fe_3O_4 MNPs to the hydrogel, these MNPs are aggregated in the hydrogel substrate due to their superparamagnetic properties (Fig. 6d). It can be realized from Fig. 6e that GO has a layered structure that creates thin and homogeneous graphite layers [61]. Fig. 6f demonstrates the κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite's morphology, driven by a combination of GO and κ -car-g-PAAm hydrogel. The layered structure of GO is visible in the

hydrogel substrate whose morphology was explained.

3.2.4. VSM

The VSM analysis of the sample is executed to measure the magnetic properties of the prepared nanocomposite. According to the M–H curves shown in Fig. 7, the magnetic coercivity and magnetic remanence of Fe_3O_4 MNPs, κ -car-g-PAAm@ Fe_3O_4 , and κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposites are zero. These results confirmed that three prepared samples were superparamagnetic. According to the M–H curves of the Fe_3O_4 MNPs, the magnetic saturation (M_s) value is 58.0 emu/g, and the M_s value for κ -car-g-PAAm@ Fe_3O_4 and κ -car-g-PAAm@ Fe_3O_4 -GO has decreased, respectively. The observed reduction in the M_s value is related to GO and κ -car, which are not magnetic. The M_s value of κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite has decreased compared to Fe_3O_4 MNPs because the hydrogel nanocomposite's main part consists of an organic part. However, the M_s value of κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite is enough to be collected by a magnet in experiments [62].

3.2.5. TGA analysis

Thermal stability of (a) Fe_3O_4 MNPs, (b) κ -car gum, (c) κ -car-g-PAAm, (d) κ -car-g-PAAm@ Fe_3O_4 , (e) GO, and (f) κ -car-g-PAAm@ Fe_3O_4 -GO are investigated as shown in Fig. 8, by using thermal gravimetric analysis (TGA) in the temperature range of 50 to 700 °C, nitrogen atmosphere, and 10 °C/min heating rate. In the TGA analysis of all samples, weight loss occurs below 200 °C owing to the vanishing of water molecules or entrapped solvents. Based on the TGA curve of Fe_3O_4 MNPs (Fig. 8a), 5–6% weight loss is interrelated to deoxidation and dehydration. The thermal behavior of κ -car gum has three stages (Fig. 8b). In the first stage, about 10% of κ -car degradation is related to water evaporation from the κ -car gum. The second weight loss of κ -car gum occurs with a steep slope up to 300 °C, and then a slow slope weight loss was observed. This weight loss (ca. 50%) relates to separating κ -car gum's functional groups such as –OH and –SO₃. In the third stage, the weight loss continues with a lower slope than in the previous stage due to the destruction or decay of C–O–C bonds and, finally, the breaking of the polymer chains from each other. The κ -car-g-PAAm, unlike κ -car, has

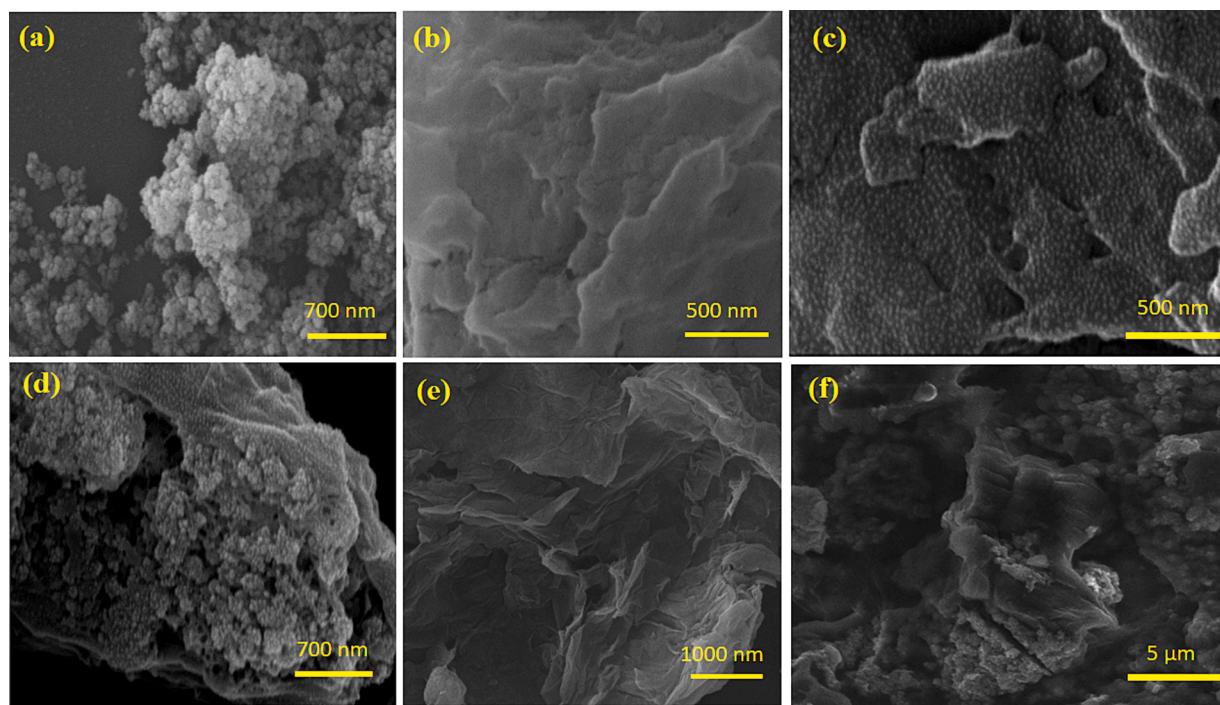


Fig. 6. SEM images of (a) Fe_3O_4 MNPs, (b) κ -car gum, (c) κ -car-g-PAAm, (d) κ -car-g-PAAm@ Fe_3O_4 , (e) GO, (f) κ -car-g-PAAm@ Fe_3O_4 -GO.

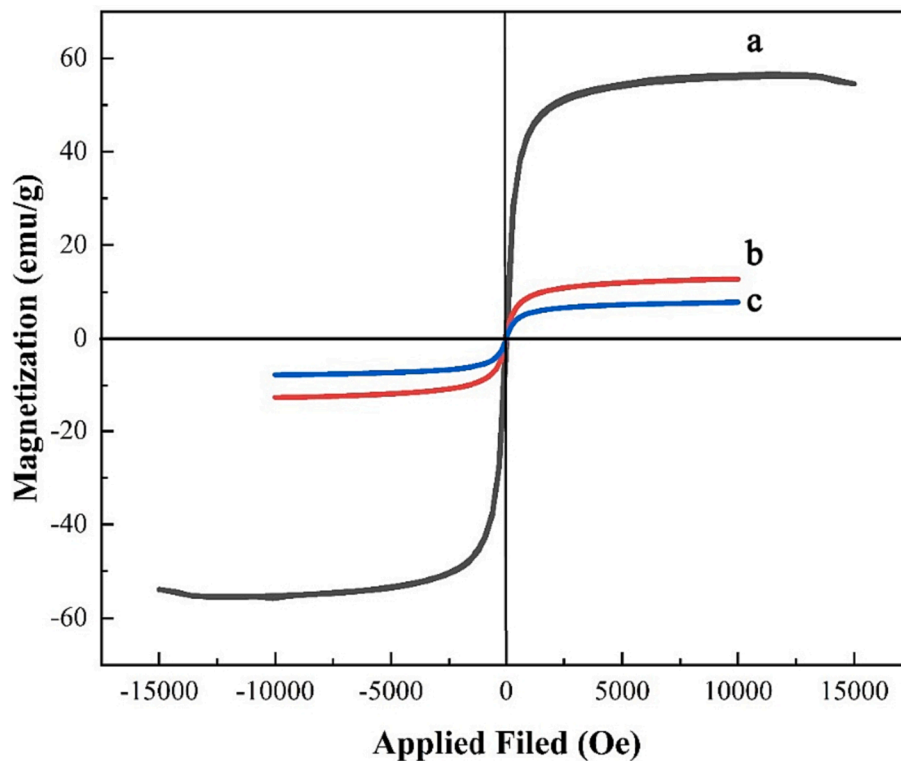


Fig. 7. The VSM curves of (a) Fe_3O_4 MNPs, (b) $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4$, (c) $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4\text{-GO}$.

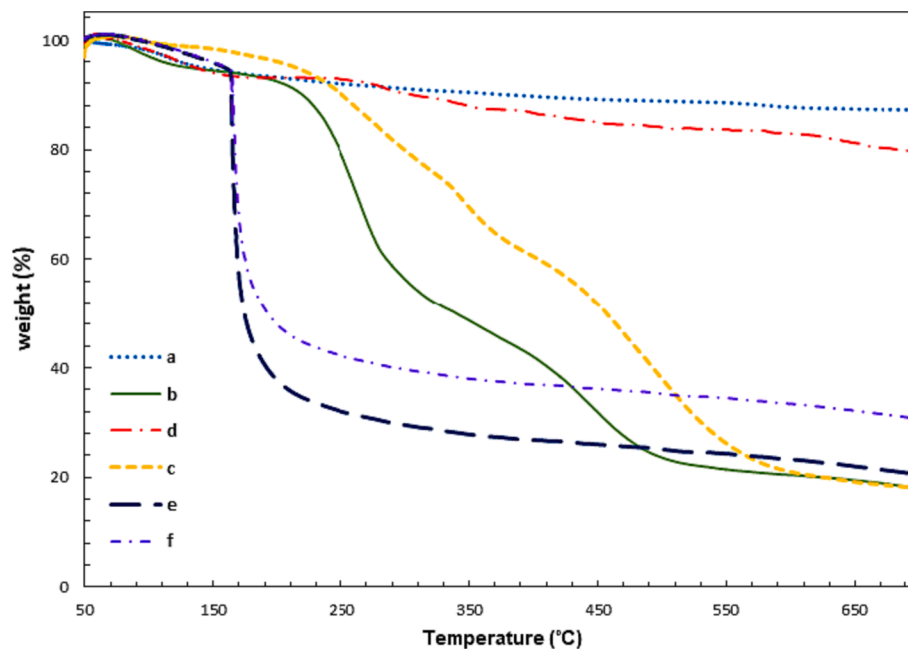


Fig. 8. TGA curves of (a) Fe_3O_4 MNPs, (b) $\kappa\text{-car}$ gum, (c) $\kappa\text{-car-g-PAAm}$, (d) $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4$, (e) GO, (f) $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4\text{-GO}$.

lost only 10% of its weight due to the separation of many functional groups, synthetic polymer, and natural polymer up to 300° (degradation of PAAm and $\kappa\text{-car}$). Finally, the destruction of the synthetic and natural polymer chains, both organic components, shows that $\kappa\text{-car-g-PAAm}$ does not provide good temperature resistance (Fig. 8c). However, as shown in Fig. 8d, about 83% of the weight of $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4$ was preserved due to the integration of the Fe_3O_4 MNPs in the hydrogel substrate as a mineral material that exhibits very high thermal resistance.

In comparison, only 30% of the weight was preserved for $\kappa\text{-car-g-PAAm}$. The TGA curve of GO (Fig. 8e), a 65% weight loss in the $400\text{--}180^\circ\text{C}$ range, can be attributed to the destruction of numerous GO's C = O and OH groups. There is also a slight weight loss between 400°C and 700°C , which can be interrelated to removing the more stable oxygenated functional group and the carbon skeleton. However in the final composition (Fig. 8f), the temperature resistance has increased compared to GO, $\kappa\text{-car}$, and $\kappa\text{-car-g-PAAm}$ due to the presence of Fe_3O_4 MNPs in the nanocomposite[63]. The $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4\text{-GO}$

nanocomposite showed char yield of about 31 % up to 700 °C.

3.2.6. XRD analysis

XRD is applied to examine the crystalline nature and particle size. According to the information reported in the articles, the X-ray diffraction pattern of Fe_3O_4 MNPs is shown in Fig. 9. The XRD pattern of the κ -car gum in Fig. 9a shows that the crystalline structure of κ -car gum appertains to the helice's oriented packing. Nevertheless, there were no sharp peaks observed in the XRD pattern, indicating that the profile arose as an amorphous compound's signature, demonstrating two peaks at $2\theta = 28.53^\circ$ and a feeble peak at ca. $2\theta = 8^\circ$ [64,65]. Fig. 9b exhibits the structure of Fe_3O_4 MNPs. The peaks show a crystalline structure at $2\theta = 30.33^\circ, 35.73^\circ, 37.38^\circ, 43.43^\circ, 53.89^\circ, 57.45^\circ, 63.10^\circ$ which are assigned to the crystal planes of (022), (131), (222), (040), (242), (151), and (044) with 96–900-6248 JCPDS standard card [66]. In the next step, the pattern of Fig. 9c is related to κ -car-g-PAAm. The crystallinity of AAm demonstrated sharp peaks at $2\theta = 11.99^\circ, 24.08^\circ$, and 28.58° . It can be assumed that the polysaccharide was modified considerably, resulting in molecular association alterations in the copolymers compared with κ -car and AAm. Due to previous studies, the XRD pattern of PAAm is amorphous, with a feeble and broad peak, which is covered by an intense bump of κ -car gum in the copolymers [65,67].

Furthermore, in Fig. 9d, in addition to the previously discussed bumps, the peaks related to the Fe_3O_4 MNPs are also detected in the XRD of the κ -car-g-PAAm@ Fe_3O_4 composite. However, the peak intensities are low while the Fe_3O_4 MNPs are placed in the polymeric substrate. The pattern in Fig. 9e corresponds to the GO crystalline structure. According to this literature, layered graphite has a strong, distinctive peak at $2\theta = 26^\circ$ with an interlayer distance of about 0.23 nm [68]. But as shown in Fig. 9e, this peak is not seen in the XRD pattern of the prepared GO, and instead, a peak with high intensity has appeared at $2\theta = 11.39^\circ$. The distance between the layers (0.77 nm) confirms the conversion of graphite to GO and the formation of OH and COOH groups during oxidation. Nonetheless, in the final κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite (Fig. 9f), all the diffraction patterns related to the incorporated materials are shown. The diffraction pattern of the nanocomposites resulting from the interaction between κ -car-g-PAAm@ Fe_3O_4 and GO is more like to the pattern of κ -car-g-PAAm@ Fe_3O_4 , which can be due to the excellent dispersion of GO in the κ -car-g-PAAm@ Fe_3O_4 polymeric network [62,69].

3.2.7. BET

The BET is a conventional technique to describe the physical adsorption–desorption of nitrogen molecules on the surface of the solid

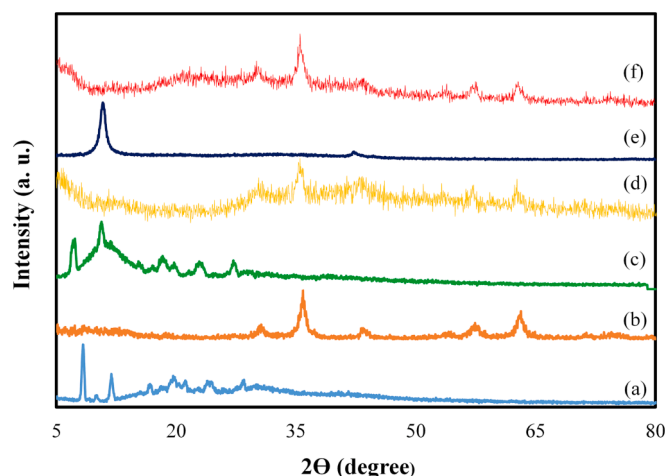


Fig. 9. The XRD patterns of (a) κ -car gum, (b) Fe_3O_4 MNPs, (c) κ -car-g-PAAm, (d) κ -car-g-PAAm@ Fe_3O_4 , (e) GO, (f) κ -car-g-PAAm@ Fe_3O_4 -GO.

materials and is utilized as a vital analytical technique for measuring the specific surface area, yet providing the surface behavior and porosity type. The N_2 adsorption–desorption analysis has been performed on GO, κ -car-g-PAAm@ Fe_3O_4 -GO, and κ -car-g-PAAm@ Fe_3O_4 (Fig. 10a–c). Based on the hysteresis loops, all samples have type IV curves, representing mesoporosity. The specific surface area was calculated through the BET method, while the samples' pore volume and pore size were measured using the BJH method, as presented in Table 3.

The prepared GO has a surface area of $9.9045 \text{ m}^2/\text{g}$, which is in relatively respectable agreement with the ideals described in the articles [70]. This surface area amount can lead to a growth in the interaction between pollutant and adsorbent molecules. The specific surface area for κ -car-g-PAAm@ Fe_3O_4 was decreased to $1.0876 \text{ m}^2/\text{g}$, which can be qualified to the graft copolymerization and cross-linking formation in the sample's structure. However, for the κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite, an increase in the surface area was observed by GO addition [62].

3.2.8. Atomic force microscopy

The images obtained from the atomic force microscope are provided to identify the recesses and protrusions of nanostructured materials' surfaces. With the help of AFM analysis, materials' structural and physical characteristics, such as roughness, hardness, particle size topography, and mechanical properties, like wear resistance, were inspected. AFM analysis was applied to investigate the topology of the surface, viz., the number of folds, wrinkles, and roughness. The 3D images resulting from this AFM analysis for GO and κ -car-g-PAAm@ Fe_3O_4 -GO can be seen in Fig. 11. The surface morphology was investigated by imaging the sample's $5 \mu\text{m} \times 5 \mu\text{m}$ area with the probe of the AFM device. In AFM analysis, it should be considered that the darker points represent the recesses and the lighter points the protrusions. Fig. 11a shows protrusions that designate the occurrence of oxygen on the outward ends of the GO structure. Fig. 11b is assigned to the κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite, where a series of changes in the morphology of the structure is observed, which is related to the modification of GO with the hydrogen of gum grafted to PAAm. Although, according to the AFM image of κ -car-g-PAAm@ Fe_3O_4 -GO nanocomposite, the depth can increase by adding Fe_3O_4 MNPs in the structure [71].

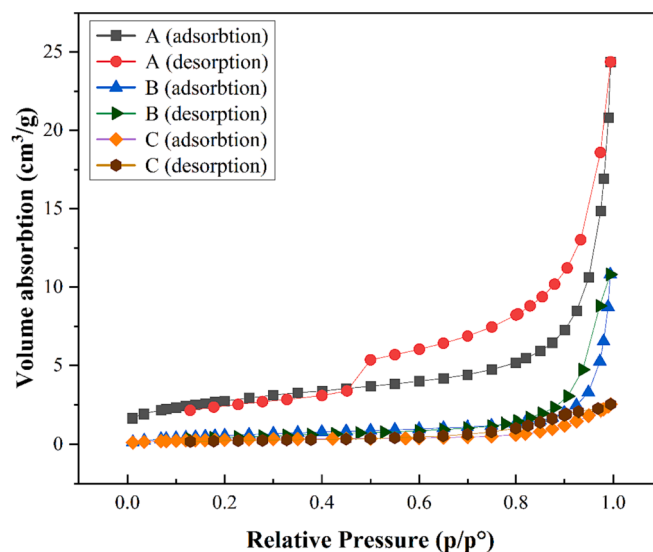


Fig. 10. The isotherm of N_2 adsorption–desorption of (a) GO, (b) κ -car-g-PAAm@ Fe_3O_4 -GO, and (c) κ -car-g-PAAm@ Fe_3O_4 .

Table 3

BET data of GO, κ -car-g-PAAm@Fe₃O₄, and κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent.

Sample	Surface area ^a (m ² ·g ⁻¹)	Pore volume ^b (cm ³ ·g ⁻¹)	Pore size ^b (nm)
GO	9.904	0.022	92.771
Car-g-PAAm@Fe ₃ O ₄	1.087	0.003	121.579
Car-g-PAAm@Fe ₃ O ₄ -GO	2.361	0.008	137.49

3.3. Effective factors on the CPF₂ adsorption by κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent

The quantity of sorbent, pH of the solution, interaction time, and CPF₂ preliminary concentration are essential parameters in CPF₂ adsorption by the prepared sorbent. Talking about these factors helps promote CPF₂ elimination effectively.

3.3.1. Zeta potential measurements

A zeta potential test was carried out to specify the prepared κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent's surface charge. The measurements were executed before the CPF₂ adsorption at an ambient temperature and in an aqueous reaction medium. Based on the outcomes summarized in Table 4, the reported zeta potential of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent at pH of 3, 5, 7, and 9 and in deionized water (DI) was

−16.1, −19.67, −23.45, and −28.98, respectively. The zeta potential's absolute value increment is attributed to elevating the pH value since the deprotonation of OH and COOH in the nano-sorbent has occurred. Significantly, based on the substantial κ -car-g-PAAm@Fe₃O₄-GO nano-adsorbent's negative surface charge, the aggregation of the nano-adsorbent particles at different pH amounts does not happen.

3.3.2. pH of medium

The solution pH was experimented on between 4 and 9 to explore the solution pH impact on the adsorption efficacy of CPF₂ by κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent, and the outcomes are depicted in Fig. 12a. The pH and pK_a are crucial solution chemistry data to investigate what chemical reactions can exert at specific pH levels. The solution's pH highly impacts the interactions between adsorbent and adsorbate. Based on the literature, two pK_a values have been reported

Table 4

Zeta potential data of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent at various pH.

Entry	Sample	pH	Zeta potential (mV) ^a
1	κ -car-g-PAAm@Fe ₃ O ₄ -GO	3	−16.1
2	κ -car-g-PAAm@Fe ₃ O ₄ -GO	5	−19.67
3	κ -car-g-PAAm@Fe ₃ O ₄ -GO	7	−23.45
4	κ -car-g-PAAm@Fe ₃ O ₄ -GO	9	−28.98

^a The experiment was exerted at ambient temperature and DI water solution.

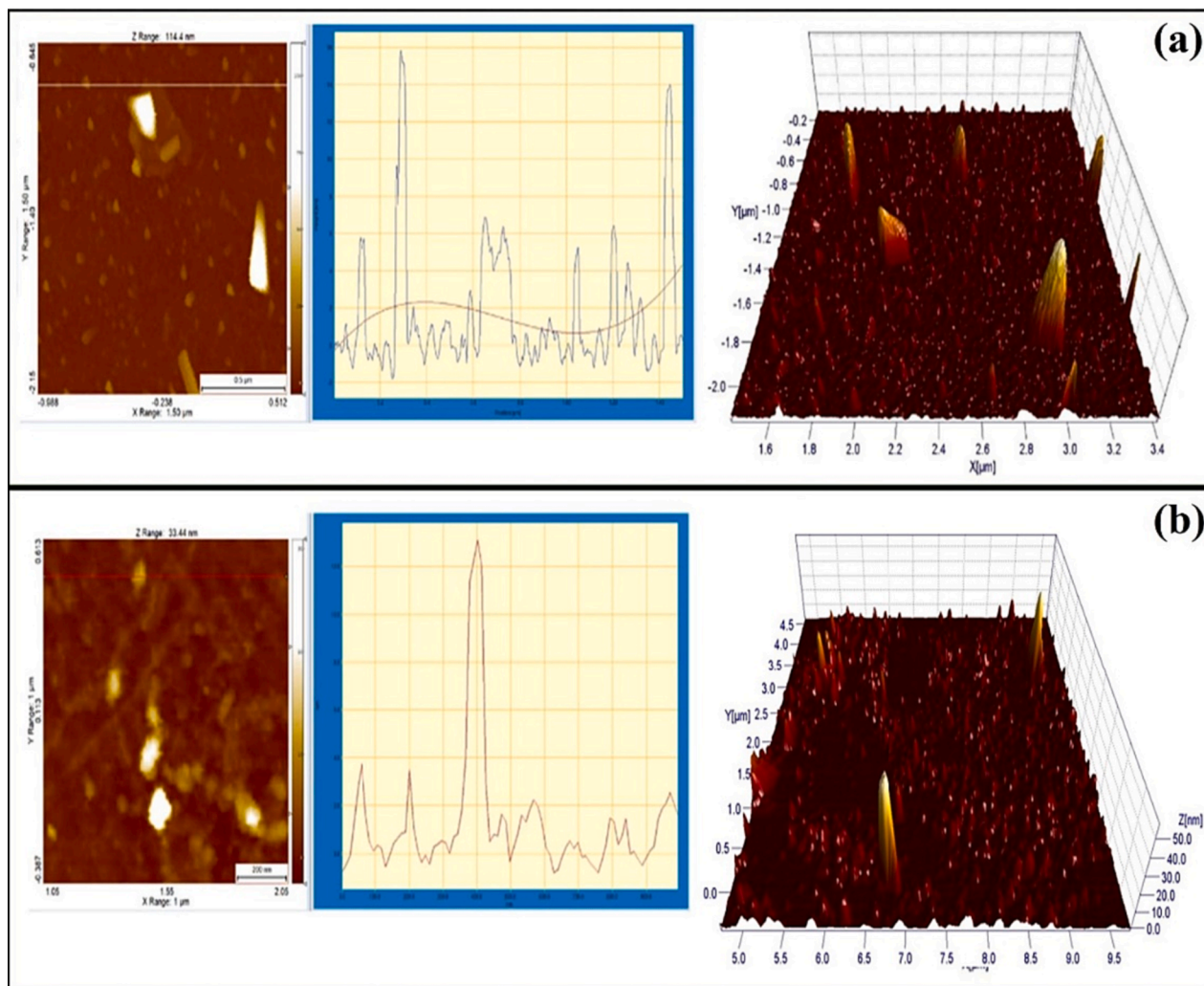


Fig. 11. Topographic images obtained from AFM analysis (a) GO, (b) κ -car-g-PAAm@Fe₃O₄-GO.

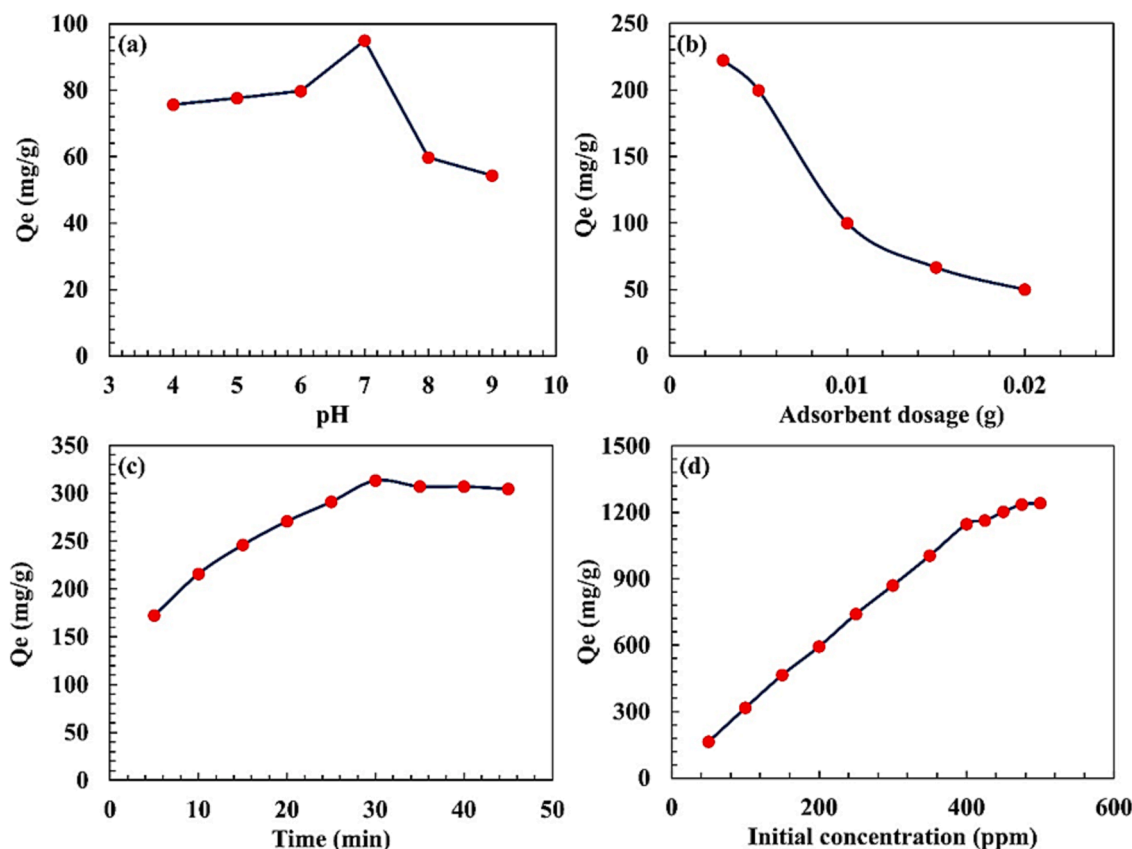


Fig. 12. (a) Effect of solution pH (b) adsorbent dosage (c) contact time and (d) initial concentration.

for CPFX at 5.90 and 8.89, attributing to the NH_2 and COOH [72]. Under the pH of 5, the CPFX renders a cationic structure due to the amine group protonation. Between 5.9 and 8.9 pH, CPFX shows zwitterionic features. However, it has an anionic structure at $\text{pH} > 8$, according to carboxylic acid's deprotonation. The adsorption capacity of CPFX antibiotic first grows with growing pH from 4 to 7 and then diminutions with growing pH. The maximum sorption capacity for CPFX antibiotic was observed at neutral pH (7), equal to 94.94 mg/g. The reason relates to the positive charge of the adsorbent surfaces in acidic pH. On the other hand, the CPFX antibiotic also exists in cationic form (NH_2^+) in acidic pHs. Therefore, the adsorption process between the positive surfaces of the adsorbent and the cationic CPFX is not favorable due to electrostatic repulsion. Also, owing to the deprotonation of the hydroxyl group in the adsorbent structure at alkaline pH and the presence of the anionic species of CPFX antibiotic, there is an electrostatic repulsion between CPFX and sorbent molecules, important to an adsorption decrease. Hydrogen bonds and electrostatic interactions between CPFX and sorbent are the main factors in CPFX antibiotic binding to $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4\text{-GO}$ adsorbent. Therefore, at neutral pH, where the concentration of H^+ and OH^- ions is at its lowest, the pressure of these ions on the interaction between the active sites of the adsorbent and CPFX antibiotic molecules is reduced. This way, the interactions between the adsorbent and the CPFX pollutant are observed at this pH. At this stage, other effective factors on adsorption, such as the quantity of adsorbent, interaction time, and preliminary concentration of the solution containing CPFX antibiotics, were considered constant. The amount of 10.0 mL of the solution containing the CPFX antibiotic in different pHs (4,5,6,7,8,9) in 0.01 g of adsorbent in 30 min, and the initial concentration of 100.0 ppm on a stirrer at room temperature, stirred. Then, the final solution concentration was measured and recorded using a UV–vis device.

3.3.3. Adsorbent quantity

At this stage, the outcome of the adsorbent's quantity on the sorption capacity was assessed, and the related diagram is shown in Fig. 12b. By reducing the quantity of sorbent, the sorption capacity for the CPFX antibiotic increases. Generally, this increase in adsorption capacity is due to the available contaminant amount increment for adsorption on the sorbent. However, the increasing amount of adsorbent, according to more aggregation of adsorbent molecules and enhancement of interaction between adsorbent molecules, causes a decrease in the surface active sites needed to carry out the adsorption process. Therefore, 0.003 g of adsorbent has the uppermost sorption capacity of 222.267 mg/g and the highest adsorption percentage of 99.8% for the CPFX antibiotic. The increment in the adsorption amount is proportional to the adsorption sites and their availability that are exposed to the pollutant solution. In this optimization stage, different amounts (0.003, 0.005, 0.01, 0.015, 0.02) of the adsorbent in the 10.0 mL of a solution containing the CPFX antibiotic at a constant initial concentration of 100.0 ppm and the ideal pH conditions, which is 7 for the CPFX antibiotic from the previous stage was stirred for 30 min at 25 °C. Finally, the final concentration of the solution was calculated [73].

3.3.4. Interaction time

The interaction time is a main parameter in the adsorption with proper efficiency, identifying the binding speed and discovery of the optimal time for the broad removal of pollutants such as antibiotics. The effect of interaction time (5–45 min) on the pollutant adsorption at optimal pH and the sorbent quantity was executed to evaluate the adsorption efficiency of the CPFX by the $\kappa\text{-car-g-PAAm@Fe}_3\text{O}_4\text{-GO}$ adsorbent. As depicted in Fig. 12c, by increasing the interaction time of the adsorbent with the CPFX antibiotic solution, the adsorption capacity increased by 313.3 mg/g in the first 30 min. However, after 30 min, no noticeable adsorption capacity increase was detected with the growth in

the interaction time. In the time optimization step, 10.0 mL of the solution containing CPFX antibiotic at diverse times (5–45 min), with the optimal amount of κ -car-g-PAAm@Fe₃O₄-GO adsorbent (0.003 g) at neutral pH and constant concentration of 100.0 ppm were considered. The optimum time for CPFX adsorption was 30 min. Based on the obtained results, initially, there were more active sites on the adsorbent to interact with pollutant molecules. However, over time these active sites become saturated, and in this way, no noteworthy growth in sorption capacity is detected [73].

3.3.5. Initial pollutant concentration

The adsorption efficiency was measured under optimal conditions to examine the preliminary concentration impact on the adsorption rate of the CPFX antibiotic. Fig. 12d displays a direct association between the initial CPFX concentration and the adsorption capacity. In the last optimization step, the CPFX initial concentration was acquired in such a way that 10.0 mL of the solution containing CPFX in different concentrations of 50, 100, 150, 200, 250, 300, 350, and 400 ppm with an adsorption amount of 0.003 g at the adjusted neutral pH, CPFX antibiotic was stirred for 30 min at 25 °C on a magnetic heater. Finally, the concentration was measured [73]. Therefore, by the CPFX solution concentration increment from 50 mg/L to 400 mg/L, the adsorption capacity of CPFX antibiotic by κ -car-g-PAAm@Fe₃O₄-GO adsorbent reached from 163.733 mg/L to 1146.467 mg/L.

3.4. Isotherm and kinetics studies

So far, the evaluation of the equilibrium adsorption of the pollutants has been described through various isotherm models, including Freundlich, Langmuir, Temkin, etc. The adsorption isotherms specify the antibiotic adsorption capacity and help identify the superficial characteristics of the prepared adsorbent. The Langmuir and Freundlich isotherms described the adsorption isotherms at equilibrium in the current work. Regarding the statistical thermodynamic laws, a monolayer pollutant sorption on the dynamic places of the sorbent surface was explained via Langmuir isotherm. This equation assumes that all of the adsorbent surface's active sites possess similar adsorption energy. On the contrary, the multilayer pollutant adsorption on the

heterogeneous sorbent surface was defined by the Freundlich equation. The Langmuir and Freundlich isotherm equations are represented in Eq.4 and Eq.5, respectively [74].

$$\frac{1}{Q_e} = \frac{1}{Q_m} + \frac{1}{K_L Q_m C_e} \quad (4)$$

$$\ln Q_e = \ln K_F + \frac{1}{n} \ln C_e \quad (5)$$

where K_L and K_F stand for the constants of Langmuir and Freundlich isotherms, respectively, both with L/mg unit. C_e (mg/L) is the concentration of the sorbate after the equilibrium. Q_e (mg/g) and Q_m (mg/g) are the amount of antibiotic contaminant at equilibrium and the highest pollutant adsorption capacity. The employed conditions for the efficient adsorption reaction were an initial concentration of 50.0–400.0 mg/L for CPFX, pH 7, adsorbent dosage (0.003 g), V (10.0 mL), contact time (30 min), T (298 K) based on which the adsorption isotherms are evaluated. Concerning Fig. 13a–b, and the reported results in Table 5, by comparing the R^2 value of 0.977 and 0.985 for the Langmuir and Freundlich models in CPFX adsorption, respectively, the Freundlich model defining a multilayer CPFX adsorption on the κ -car-g-PAAm@Fe₃O₄-GO adsorbent surface well describes the adsorption model.

Table 5

Isotherm and kinetic constants for CPFX adsorption on the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent.

Model		Parameters	CPFX
Isotherm	Freundlich	K_F (mg/g)	168.308
		N	2.282
		R^2	0.985
	Langmuir	Q_{max} (mg/g)	1000
		K_L (L/mg)	0.709
Kinetics	Pseudo-first-order	R^2	0.977
		k_1 (min ⁻¹)	0.084
		Q_e (mg/g)	226.512
	Pseudo-second-order	R^2	0.911
		k_2 (min ⁻¹)	0.0004
		Q_e (mg/g)	357.142
		R^2	0.995

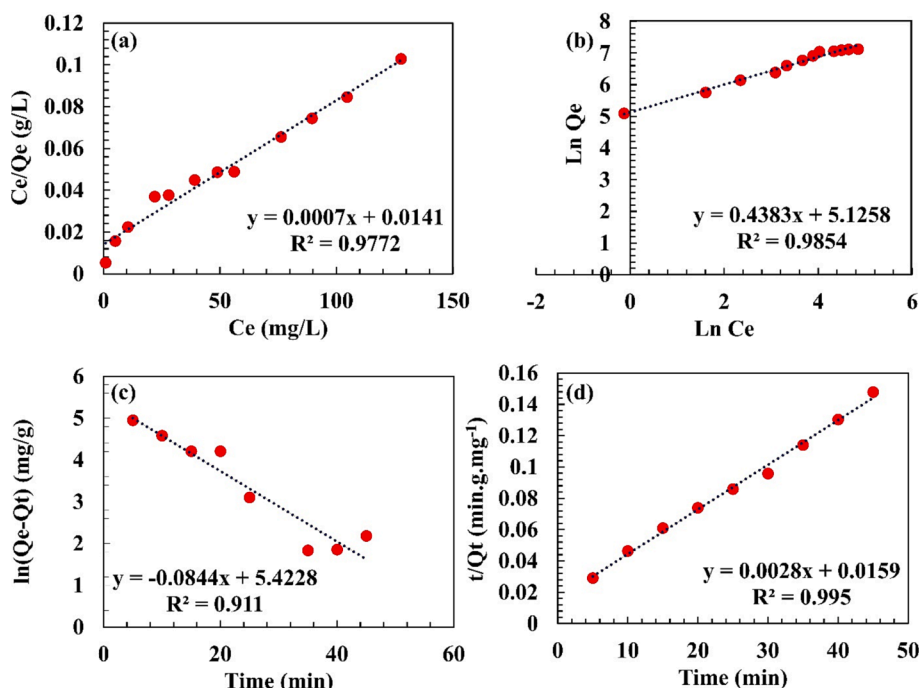


Fig. 13. (a) Langmuir and (b) Freundlich(c) pseudo-first-order and (d) pseudo-second-order models.

Generally, the adsorption kinetics process exhibited the development of the adsorption over time, evaluating the adsorption rate and mechanism data. Numerous kinetic representations are presented to define the model of the adsorption kinetics information, such as Elovich, pseudo-first-order, and pseudo-second-order. The leading kinetic models of the current work are pseudo-first-order and pseudo-second-order, as determined in Eq. (6) and Eq. (7), respectively [75].

$$\ln(Q_e - Q_t) = \ln Q_e - k_1 t \quad (6)$$

$$\frac{t}{Q_e} = \frac{1}{Q_e} t + \frac{1}{k_2 - Q_e^2} \quad (7)$$

where Q_e and q_t are the sorbed antibiotic amount with mg/g unit at equilibrium (e) and the related time (t). k_1 (1/min) and k_2 (g/mg.min) define the first-order and second-order rate constant, respectively. The adsorption tests were executed in 5 to 45 min intervals at an ambient temperature, utilizing 0.003 g of the κ -car-g-PAAm@Fe₃O₄-GO adsorbent in 10.0 mL of CPFX solution with 400.0 ppm concentration, respectively. The adsorption kinetics linear plots and the related information of CPFX antibiotic pollutants are shown in Fig. 13c-d, and Table 5, respectively. Due to the R^2 value indicating the correlation coefficient and alteration between the experimental and calculated Q_e , the pseudo-second-order models well describe the CPFX contaminant adsorption kinetics. This result proves that the adsorption procedure on the κ -car-g-PAAm@Fe₃O₄-GO adsorbent surface is controlled by chemical adsorption. In addition, according to the data in Table 5, the Q_e values obtained for the second-order pseudo-kinetic model are closer to the equilibrium Q_e than the Q_e values obtained from the first-order pseudo-kinetic model.

A brief comparison has been reported in Table 6 between the CPFX $Q_{\max}$ of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent and other previously studied adsorbents considering the pH and the reaction time. According to the presented data, the $Q_{\max}$ of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent was more enhanced than the reported adsorbents, ascribed to the improved affinity of functional groups of the prepared nanoadsorbent, viz., hydroxyl and carboxyl, and epoxy, interacting with the CPFX pollutant properly. Despite the low-priced bioadsorbents (in most cases, the residues of the agricultural materials) than κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent, the efficacy of the CPFX elimination by κ -car-g-PAAm@Fe₃O₄-GO is higher.

3.5. Recyclability of κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent in adsorption procedure

Recycling and reusing the nanoadsorbent is essential due to its

Table 6

Comparing the $Q_{\max}$ of κ -car-g-PAAm@Fe₃O₄-GO nanosorbent to previous reports.

Entry	Adsorbents	$Q_{\max}$ (mg/g)	pH	Time (min)	Ref.
1	GO	379	5	240	[76]
2	MCGO ^a	282.9	5	360	[77]
3	MGO@PANI ^b	106.38	6	30	[78]
4	MGOCP ^c	283.44	6	60	[79]
5	Zn-GO@SA ^d	134.85	7	600	[80]
6	GO/O-CNTs ^e	512	6	240	[81]
7	GO/AB ^f	62.384	7	420	[82]
8	κ -car-g-PAAm@Fe ₃ O ₄ -GO	1146.467	7	30	This work

^a Magnetic chitosan grafted graphene oxide nanocomposite,

^b Polyaniline-magnetic graphene oxide nanocomposite,

^c Fe₃O₄/GO/citrus peel-derived magnetic bio-nanocomposite,

^d Zinc ion cross-linked GO and sodium alginate aerogel microspheres,

^e Graphene oxide/oxidized carbon nanotubes,

^f Graphene oxide-alginate beads.

affordability, environmental protection, and saving time and energy. Adsorption-desorption experiments were accomplished for three serial cycles to approximate the recyclability and renewal of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent. CPFX was desorbed from the used nanoadsorbent (CPFX-loaded κ -car-g-PAAm@Fe₃O₄-GO) by submerging and inspiring the nanoadsorbent in EtOH at 25 °C for 4 h. Therefore, CPFX was free in the solution, and the nanoadsorbent was collected by a magnet, rinsed for numerous periods with H₂O, and dried for subsequent cycles. As can be detected in Fig. 14, the adsorption diminished from 85.98% to 81.19%, and the removal percentage diminished from 83.0% to 77.56% after three cycles. These outcomes indicate that the nanoadsorbent holds the ability to eliminate CPFX from aqueous solution after three successive cycles without a noteworthy diminution in adsorption performance.

3.6. Anticipated mechanism of adsorption

The textural characteristic, natural structure, and numerous functional groups of κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent are very crucial in the adsorption procedure of antibiotics. Hydrogel-based nanocomposite provides a large surface area, high porosity, several active groups or reactive sites (C = O and OH), and a 3D network. Notably, the solution pH, the surface charge of the sorbent, and the adsorbate's pKa can help find the solution chemistry and how it specifies the plausible mechanism that would exert [23]. CPFX rendered two pKa values at 5.9 (pKa1) and 8.9 (pKa2) [83]. The interactions between the adsorbent-adsorbate alter with the solution pH variations. The deprotonation of CPFX's carbonyl group takes place at 5.9 (pKa1). In contrast, the protonation of amine groups occurred at 8.9 (pKa2) [84]. Regarding this, the CPFX can render cationic, anionic, and zwitterion properties (Fig. 15). The negatively-charged κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent can adsorb CPFX through cation exchange by the protonated amine group [85]. According to the reported studies, the leading mechanisms of CPFX adsorption are π - π interactions, hydrogen bond network, hydrophobic interactions, electrostatic interactions, and pore diffusion [86,87]. Hydrophobic interaction forms when CPFX adsorption happens because of its low solubility [88]. The pH region of 5.90 to 8.89 is where CPFX has low solubility, performs neutral properties, and is a zwitterion. In the case of opposite charges of adsorbate and adsorbent, the electrostatic attraction comes into play. Electrostatic interactions occur once the adsorbent exhibits a negative charge around pH 1 to 5.59 and can be somewhat ongoing up to pH 8. As shown in Fig. 15, by considering the net negative charge of the sorbent in the 1.0 to 5.59 solution pH range, electrostatic interactions and cation exchange

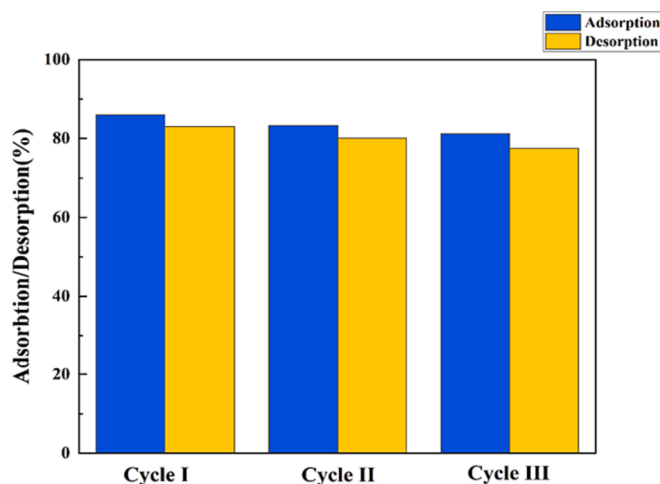


Fig. 14. The adsorption/desorption (%) of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent for CPFX removal for three consecutive runs.

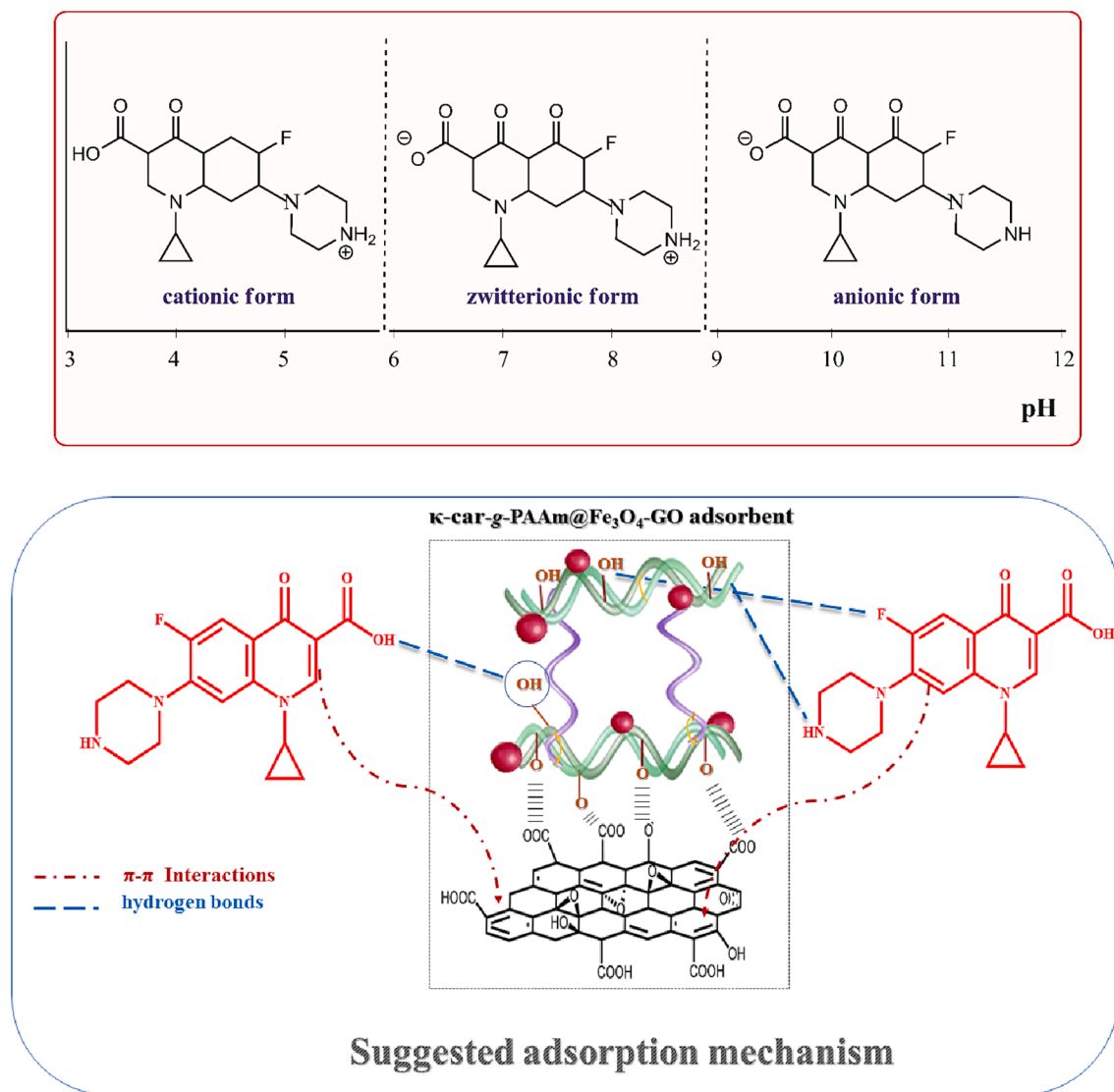


Fig. 15. Possible adsorption mechanism of the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent.

between the NH₃⁺ of CPFX's and the nanosorbent sometimes become active. Also, the capability of CPFX structure in forming π - π interactions with κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent is ascribed to electron-rich benzene ring presence. They act more potently in the vicinity of the solution's neutral pH, ranging from 5.90 to 8.89, or in the presence of CPFX as a zwitterion. Analogous pH ranges for π - π interactions enable hydrogen bond networks [72]. Nonetheless, because of the anionic form of CPFX in pH of 8.9, the leading electrostatic repulsion between the CPFX and the nanosorbent's negative charge prevailed.

4. Conclusion

The preparation of κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent was accomplished in several stages and applied for CPFX removal. The characteristics of the prepared nanoadsorbent were investigated through FTIR, EDX, TGA, SEM, VSM, BET, and XRD approaches. The most effective CPFX adsorption on the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent was carried out in the optimum adsorption reaction conditions, viz., 0.003 g adsorbent dosage, in the pH of 7 at an ambient temperature, contact time of 30 min, and 400.0 mg/L starting CPFX concentration. According to the results obtained from various analyses, the κ -car-g-PAAm@Fe₃O₄-GO nanoadsorbent exhibited good thermal stability and superparamagnetic features. The nanocomposite's XRD

pattern revealed a semicrystalline structure by adding Fe₃O₄ MNPs into the amorphous matrix of the κ -car-g-PAAm. Based on the FESEM images, a desired dispersion of the Fe₃O₄ MNPs with uniform size and shape and lamellar GO structures are detected all over the smooth κ -car-g-PAAm hydrogel surface. Also, the GO incorporation in the κ -car-g-PAAm@Fe₃O₄ magnetic adsorbent structure substantially increased the BET surface area. The pseudo-second-order kinetics (0.995) and the Freundlich isotherm (0.985) with Q_{max} of 1146.467 mg/g described the adsorption kinetics and isotherm model at room temperature and pH 7. The hydrogen bonding and π - π interactions between CPFX and functional groups on the adsorbent are the leading interactions to eliminate CPFX from water. Moreover, the applied nanocomposite in CPFX removal was simply unglued from the mixture and reprocessed for three consecutive cycles with a negligible loss in adsorption performance.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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